

M1 - BIM

Algorithms on trees and graphs in bioinformatics

Cours 1: Introduction

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The plan of today's lecture

1. What is **bioinformatics**: 2 examples of the interaction between bioinformatics and experimental biology.
2. Basic notions in biology: the orders of magnitude.
3. Various computational questions illustrating the fundamental concept of evolution:
 - I. Genome assembly and metagenomics
 - II. Search for biological signals in sequences
 - III. Sequence alignment and pointwise mutations
 - IV. Phylogeny – the reconstruction of the history of species
 - V. Structural mutations and genome rearrangements
 - VI. Prediction of protein structure – the Nobel Prize 2024
The Conversation : "The Prix Nobel de Chimie 2024"
 - VII. Systems biology and the generation of omics data



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What is bioinformatics?

Bioinformatics is usually defined as the analysis, prediction, and modeling of **biological data** using computers.

It develops novel **methods** rooted in mathematics, computer science, and physics to answer biological questions.

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Two interfaces with biology

Quantitative biology approaches study living systems with experiments, theory, or a combination of both, under the common denominator that results are not descriptive but relate to precise measurements. It includes **high-throughput techniques and the analysis of their resulting data, quantitative cell biology, theoretical modeling of experimental data and mathematical/computational approaches to data analysis.**

Computational biology develops and applies computational methods to predict and understand biological structures/processes, taken in a broader sense, and infer quantities measuring these structures/processes. **It generates new biological data coming from *in silico* computation**, and allows analyzing quantities and processes that are not directly accessible for biological analysis.

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About the course

1. Introduction to several problems in bioinformatics
 - genome sequencing, genome rearrangements
 - phylogeny of species and molecular evolution
 - molecular interaction and biological networks
2. The construction of **phylogenetic trees**
3. Algorithms for studying **genome rearrangements**
- 4-5. Reconstruction of **biological networks**
 - Random graphs and "small world" graphs
 - Properties of large graphs

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In bioinformatics we aim:

- To study genome organisation
- To explain and quantify biological processes

⇒ Bioinformatics tries to understand or discover biological mechanisms

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1. An example of a search for a mechanism: **networks of proteins in chemotaxis**

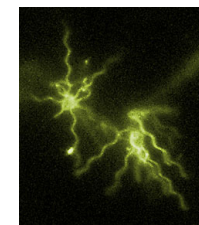
⇒ Algorithms +
Combinatorial structures +
physics

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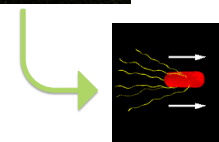
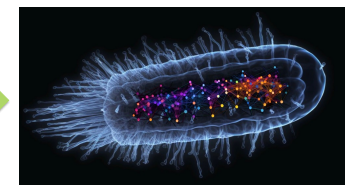
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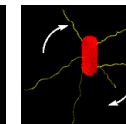
Chemotaxis



A network of molecular interactions



nager...



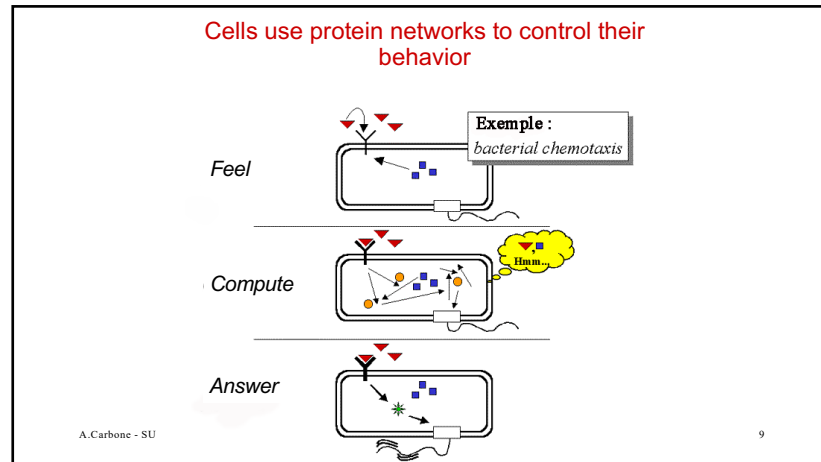
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A systemic vision of
the bacterial behavior

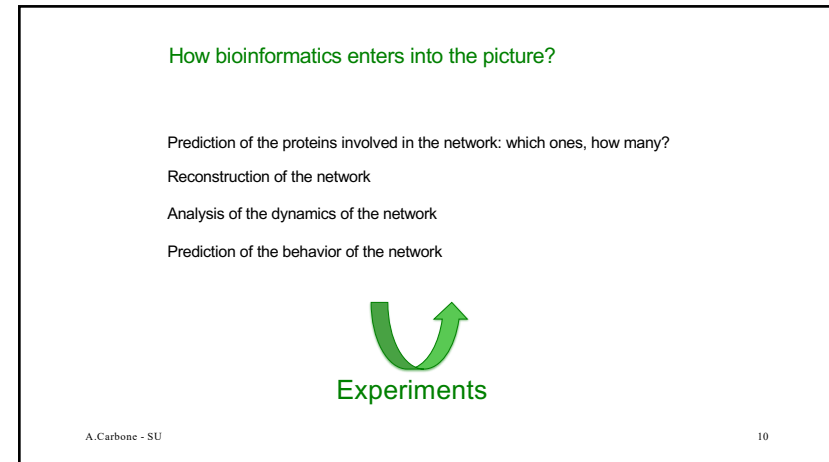
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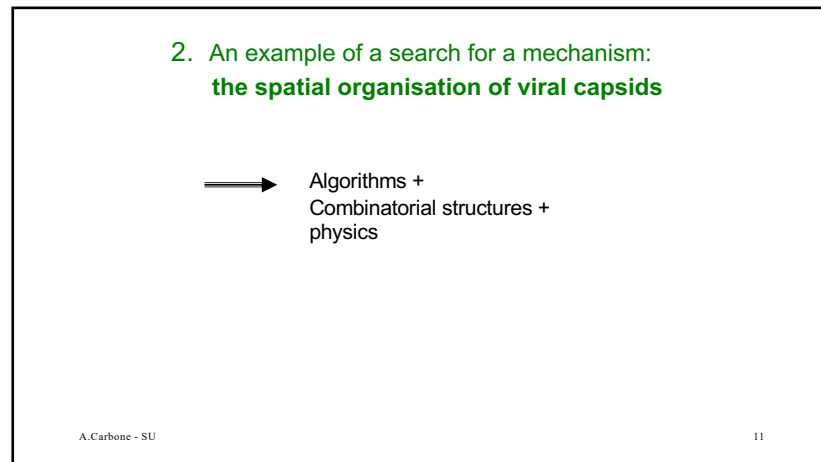
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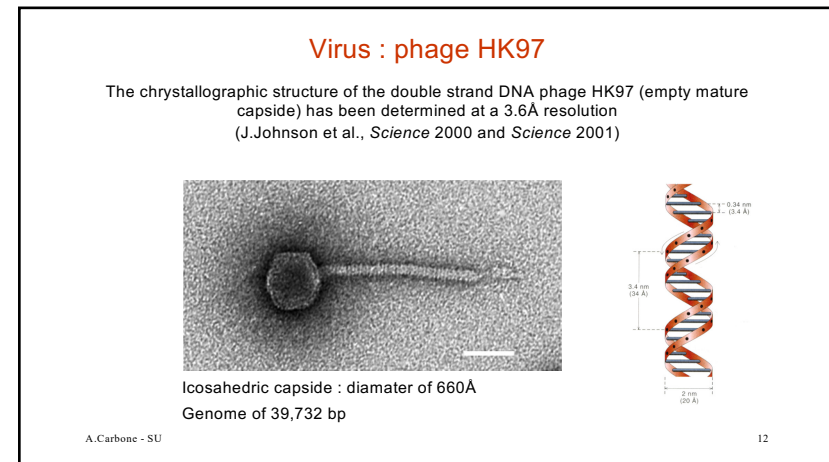
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Topologically linked proteins,
rotational subunits and local re-folding

Pentamers
(12 total)

Hexamers
(60 total)

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J.Johnson et al., *Science* 2000

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Intermediary conformations:
from "Pro-head" to "Head"

by cryoelectron microscopy (cryo-EM):

gp5 - head protein
gp4 - protease
gp3 - portal

spherical shape

icosahedral shape

assembly

cleavage

expansion 5nm

cross-linking

Prohead I

Prohead II

Head I

Head II

J.Johnson et al., *Science* 2001

What is the mechanism allowing capsid formation?

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How structural bioinformatics enters into the picture?

- Modeling of the transition between the second and the fourth capsid states
- Reconstruction of the third capsid shape
- Prediction of the stability of the intermediate structure
- Possibly: what are the missing elements in the model?

Model validation

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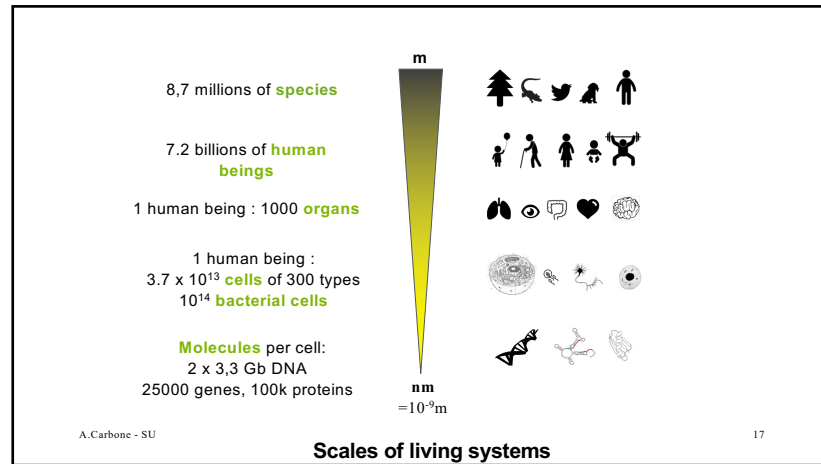
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Some important basic notions:
orders of magnitude in biology!

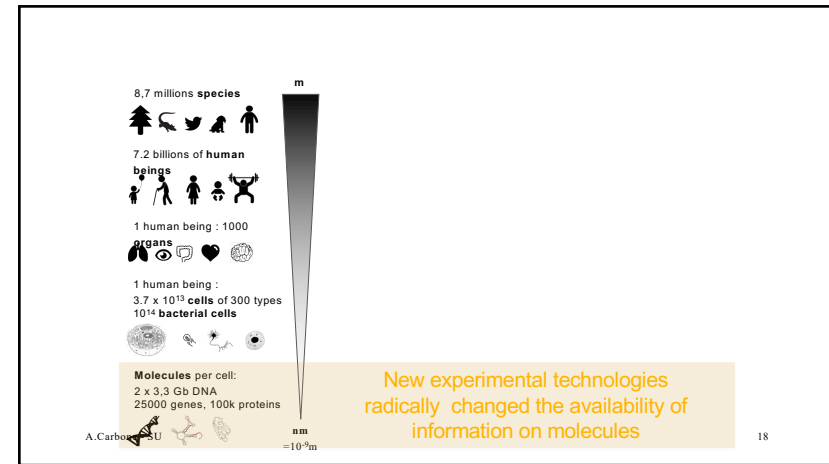
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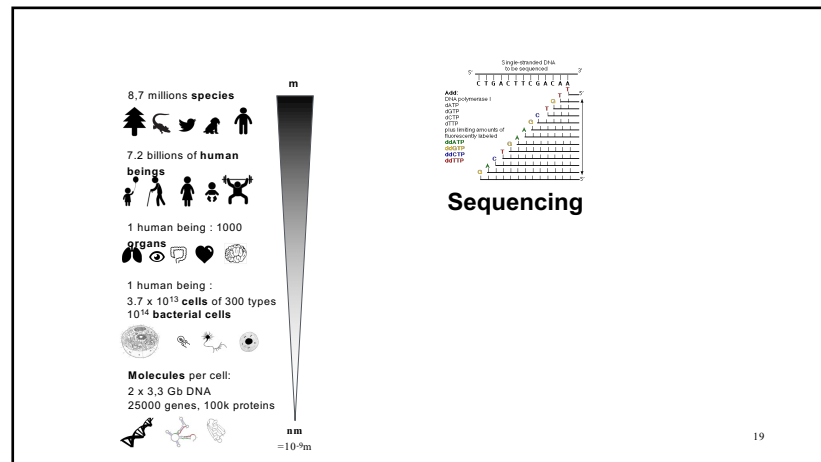
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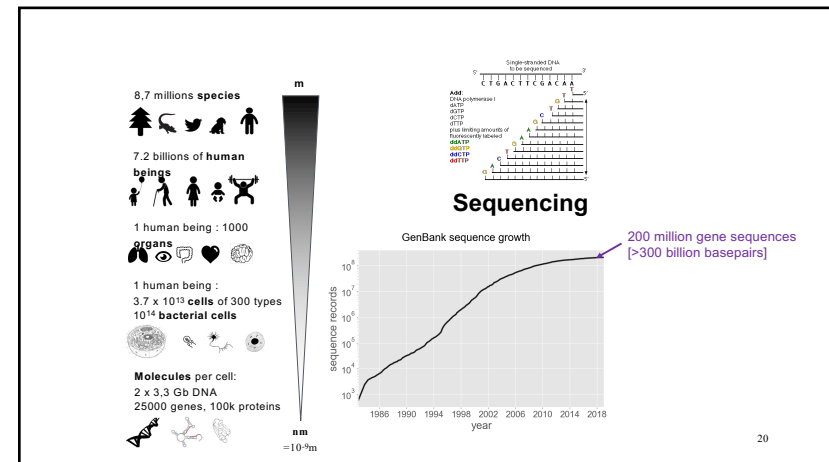
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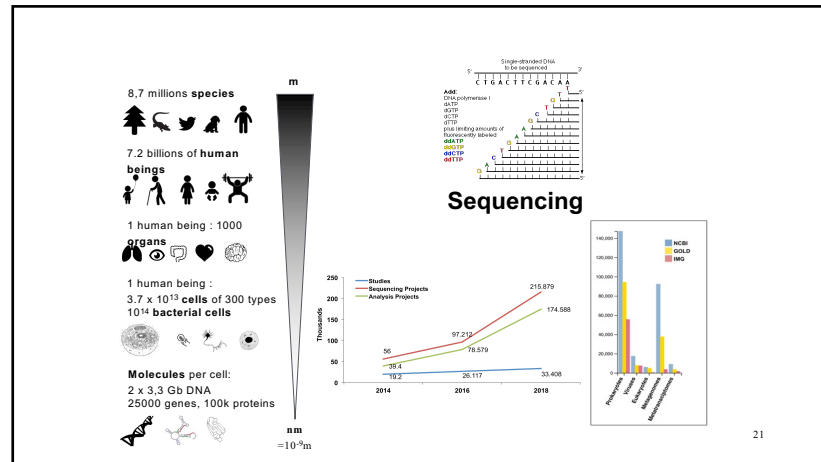
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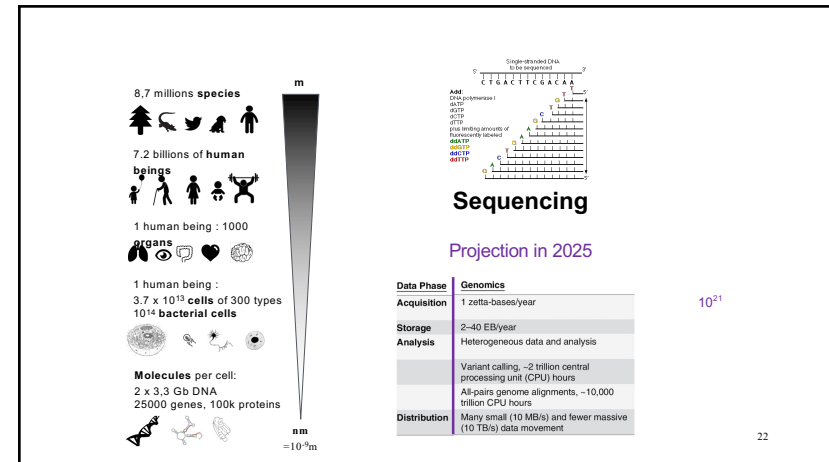
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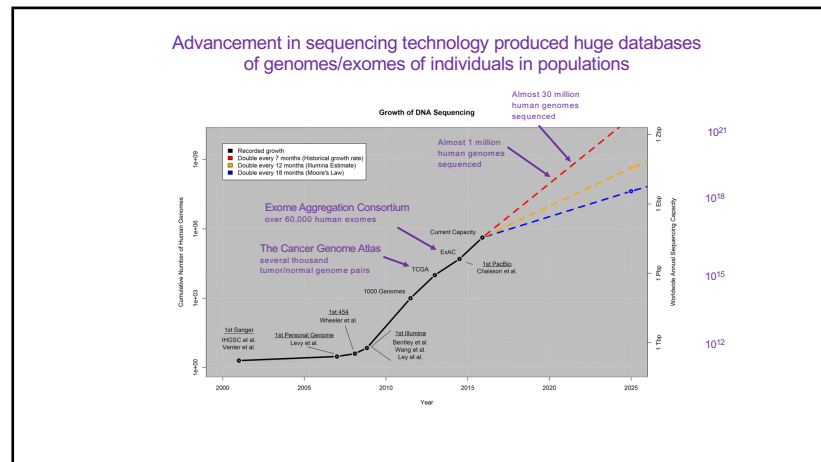
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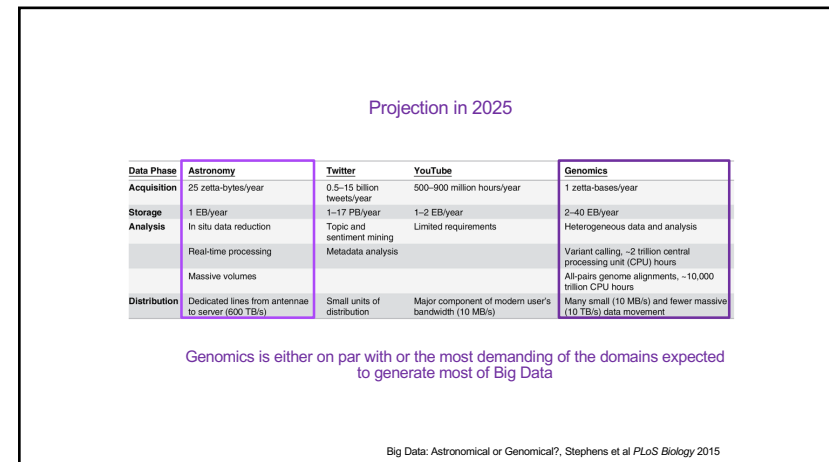
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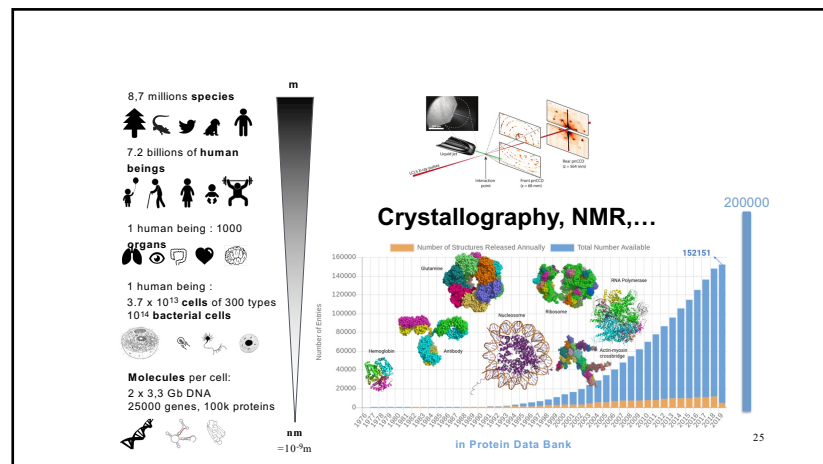
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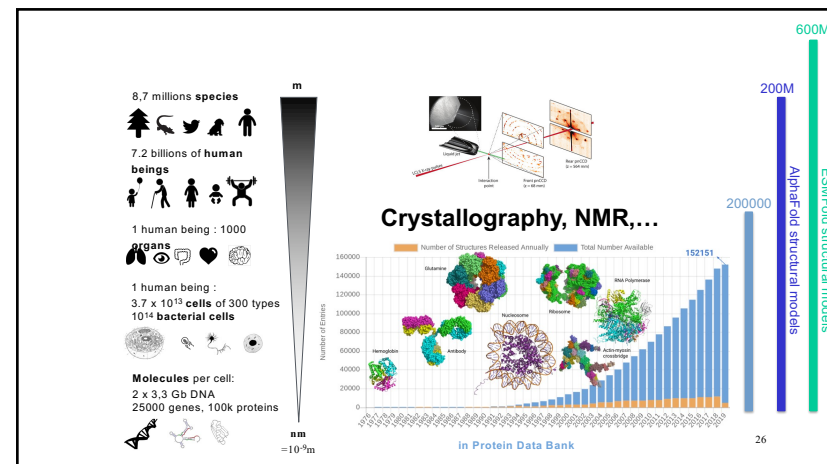
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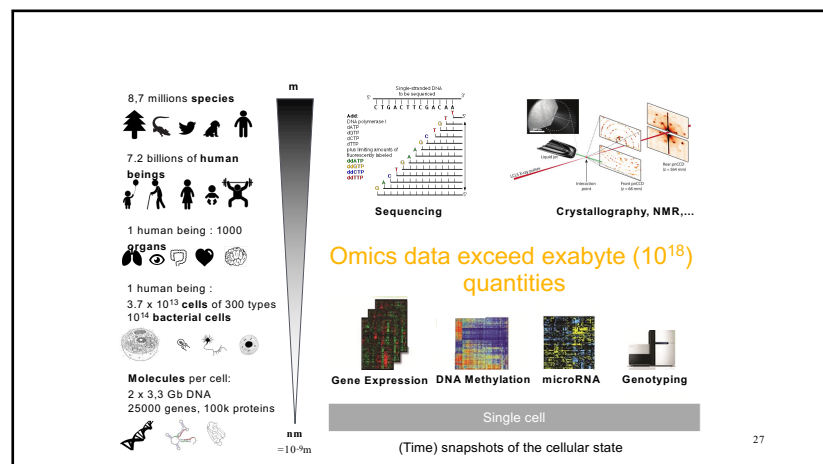
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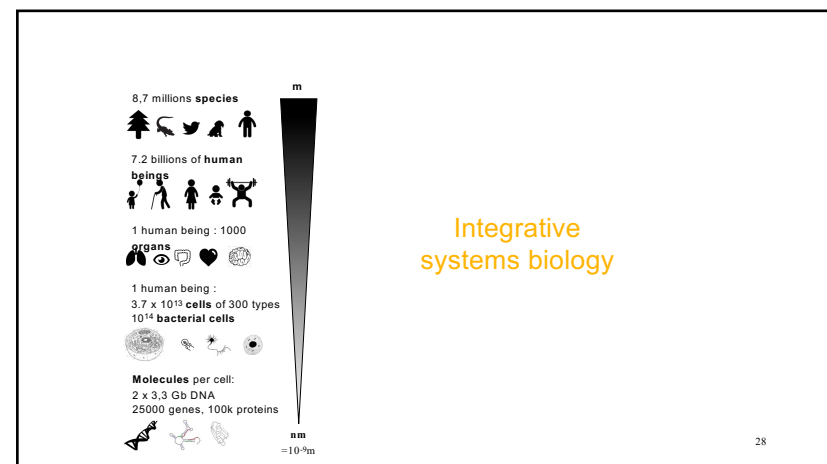
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Orders of magnitude
to be revised at home!

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Translation: redondance of the genetic code

		Second position				
		U	C	A	G	
First position (5'-end)	U	UUU phe UUC UUA leu UUG	UCU UCC UCA ser UUG	UAU tyr UAC UAA Stop UAG Stop	UGU cys UGC UGA Stop UGG trp	U C A G
	C	CUU CUC CUA leu CUG	CCU CCC CCA pro CCG	CAU his CAC CAA CAG gln	CGU CGC CGA CGG arg	U C A G
	A	AUU AUC ile AUA AUG met	ACU ACC ACA thr ACG	AAU AAC AAA AAG lys	AGU ser AGC AGA AGG arg	U C A G
	G	GUU GUC GUA val GUG	GCU GCC GCA ala GCG	GAU asp GAC GAA glu GAG	GGU GGC GGA gly GGG	U C A G

■ Initiation ■ Termination

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Table of the energy values

	kcal/mol
average kinetic energy of thermal motion per molecule at room temperature ($25^{\circ}\text{C} \approx 293^{\circ}\text{K}$)	0.9
energy of a hydrogen bond	1-5
approximate energy of covalent phosphate bond between nucleotides	50
approximate energy of $C=C$ bond	200

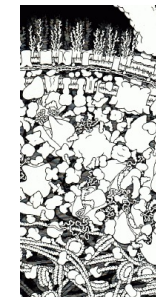
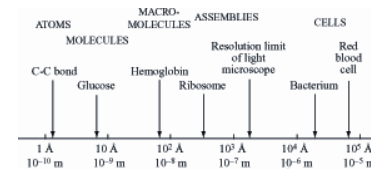
Spontaneous formation →

Note: 1 mol = quantity of substance containing 6.0221367×10^{23} molecules

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Scales - dimensions

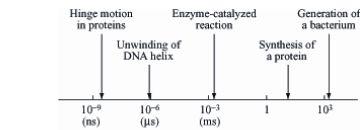


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Scales and dimensions (continued)

hydrogen atom		1.008Da
carbon C^{12}		≈ 12 Da
nitrogen, oxygen atoms		14Da, 16Da
sulfur, phosphorus atoms		32.06Da, 30.97Da
covalent bonds (distances between nuclei)	0.75-2.3Å	
hydrogen bonds	1-2Å	
van der Waals radius	2-3Å	
water molecule	3-4Å	18Da
sugars, amino-acids, nucleotides	0.5-1nm	150-500Da
globular proteins	2-10nm	$5 \cdot 10^5$ - $5 \cdot 10^6$ Da
ribosomes	25-30nm	2.5-4.5MDa
viruses	26-60nm	3-50MDa
bacteria	0.5-5µm	$5 \cdot 10^5$ - $5 \cdot 10^6$ MDa
mitochondria	2µm	10^5 - 10^6 MDa
cell nucleus	3-10µm	10^5 - $5 \cdot 10^6$ MDa
animal cells	10-30µm	$5 \cdot 10^5$ - 10^6 MDa
plant cells	10-100µm	$5 \cdot 10^5$ - 10^6 MDa
DNA in a human cell	2m	$5 \cdot 10^6$ MDa
DNA in a human body	10^9 m	200g



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Figure 15: Time of processes (in seconds).

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Some computational problems in biology !

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Computational problem I: the assembly of genomic sequences

To sequence large DNA segments, the sequence can be divided into multiple small fragments for individual sequencing. The challenge lies in reconstructing the complete sequence from these fragments.

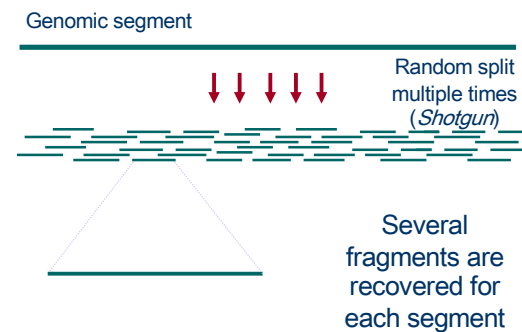
Given a set of fragments, find the sequence of minimal length which contains all members of the set of fragments.

This problem is NP-complete. There are greedy algorithms that perform the assembly in a satisfactory way and in a reasonable time. The problem is difficult because of the huge number of repeated regions.

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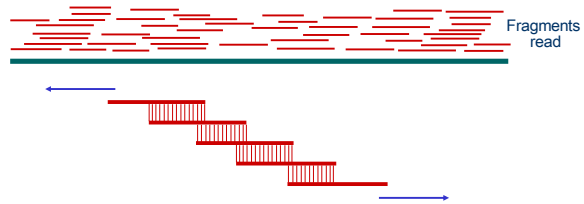
« Shotgun Sequencing »



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Assembly of the fragments



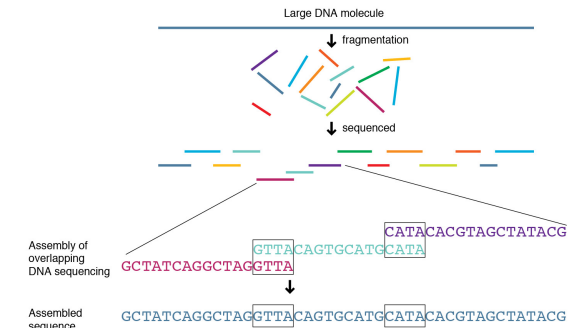
Cover the region with an abundance of ~7-fold
Overlap the fragments that have been read and extend them to reconstruct the original genomic region

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Until 1990 "shotgun fragment assembly" of the human genome has been seen as an intractable problem.

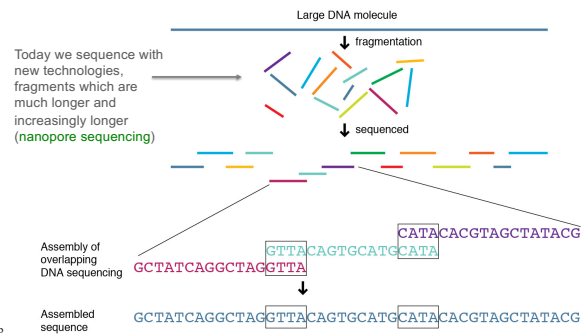


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Until 1990 "shotgun fragment assembly" of the human genome has been seen as an intractable problem.

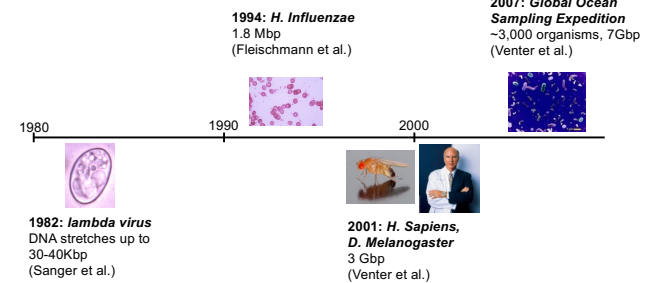


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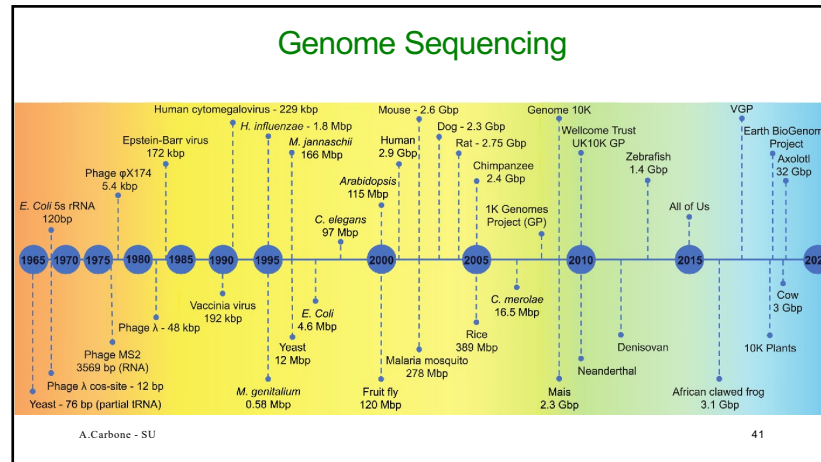
Genome Sequencing



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Difficulties in the reconstruction

Repeat/low complexity regions

Sequencing errors

Strand, usually not known

Low coverage

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Difficulties in the reconstruction

Repeat/low complexity regions

Sequencing errors

Strand, usually not known

Low coverage

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Difficulties in the reconstruction

Repeat/low complexity regions

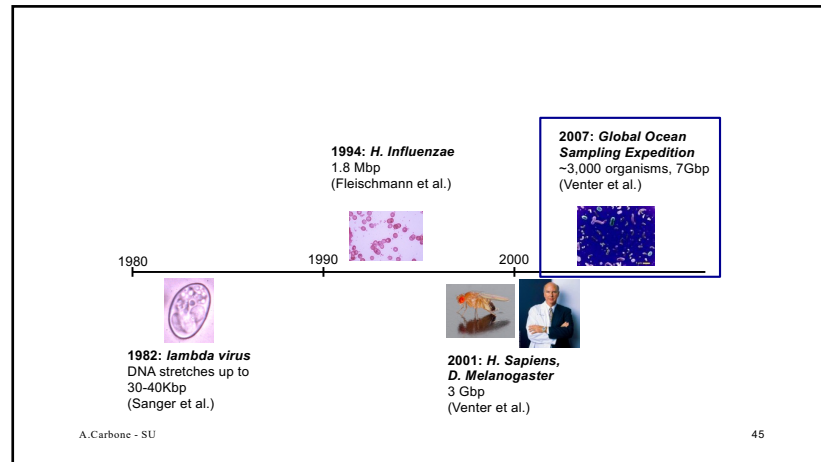
Sequencing errors

Strand, usually not known

Low coverage

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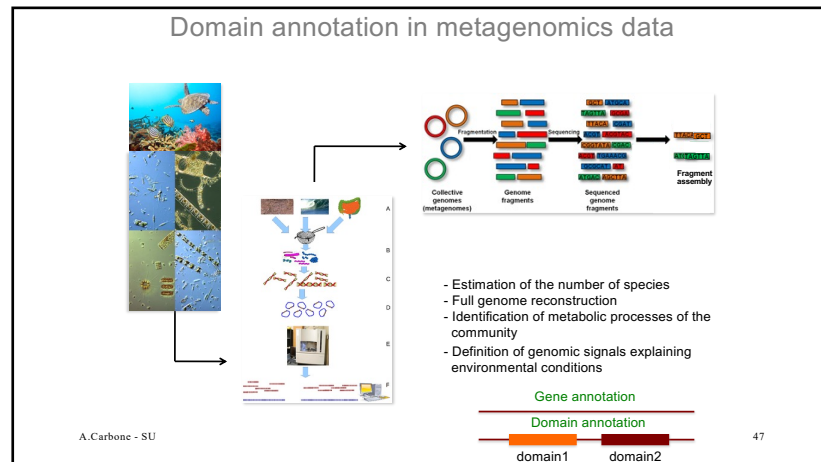
Metagenomics: reconstruction of several genomes at once and algorithmic challenges

Metagenomics = study of **metagenomes**
that is, of the set of DNA sequences that is extracted from multi-species communities in a given environment.

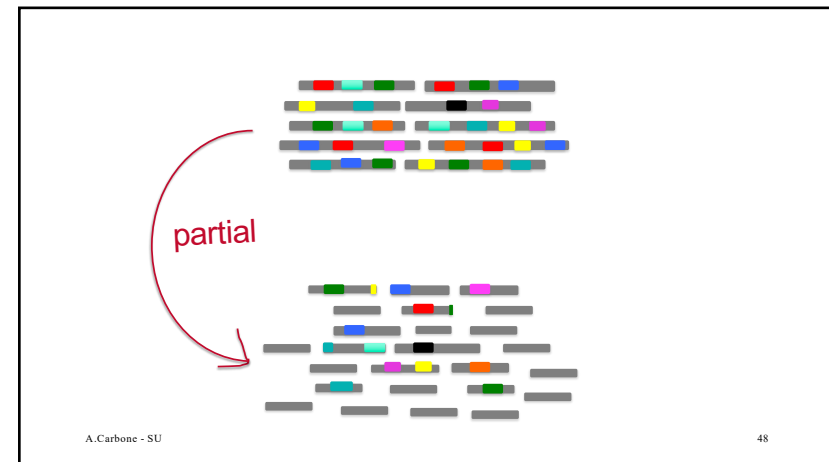
These communities are constituted by organisms that are often non cultivable in the labs. It has been estimated that only the 1% of prokaryotes within the majority of the environments can be cultivated. For the vast majority of organisms, cloning has to be realized directly on the environmental samples.

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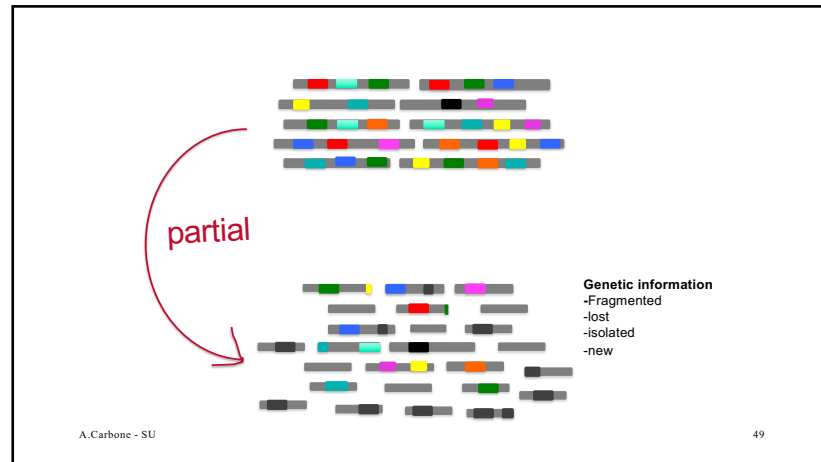
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Question addressed today: how can we find biological information in metagenomic/metatranscriptomic data?

To which species they belong to?
What functions are these sequences realizing in the environment?

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Computational problem II:
searching for signals in genomes - gene search

Given a DNA sequence, predict the position of genes (Open Reading Frames, ORF), exons and introns.

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...in prokaryotic genomes:

<http://www.tigr.org>

TIGR **tigr cmr**

Genome Data Genomes Data Data Analysis Data Analysis Data Analysis Data Analysis Data Analysis Data Analysis

Bacillus subtilis 168 Region View Display: Region Surrounding yerP

yeiP

yeiQ

Codons stop

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Gene search can be realized

ab initio, as in the method explained above,
or by exploiting information on genes already found in other genomes,
or by exploiting experimental information coming from expressed sequences.

Open problems in bioinformatics:

- the identification of small genes in large genomes
- the identification of gene sequences in viral genomes, where genes overlap
- the identification of introns/exons

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Computational problem III: pointwise mutations and sequence alignment

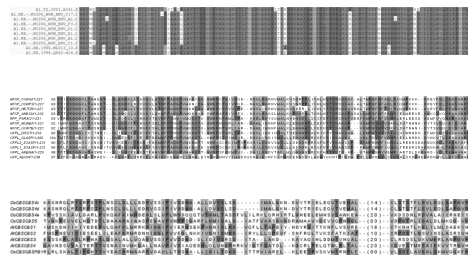
Given two (or several) DNA sequences or protein
sequences, find the best alignment

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Landscapes of homologous sequences



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A sequence **ACGTACGT** could evolve to give:

--ACG--T--A--CG--T--
ACACGGTCCT AATAATGGC C

---AC--GTA--C--G--T--
CAG--GAAGATC T TA GT TC

probability of
deletion: 0.0001
insertion: 0.001
substitution A/G, T/C: 0.00008
substitution A/C, T/G: 0.00002

Pointwise
mutation

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These two sequences have the same ancestor and we want to know the alignment which reflects their common origin (that is **ACGTACGT**)

For instance, the alignment could be

— ACAC—GGTCCTAAT— AATGGCC
CAG —GAA—G—A T—CTTAGTTC — —

Definition: an **alignment** of sequences S and T is a pair of sequences S' and T' (with gaps) such that

- (1) $|S'|=|T'|$
- (2) by eliminating gaps, we obtain S and T

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Matches, mismatches et indels

Two identical and aligned characters in an alignment form a **match**

Two different and aligned characters in an alignment form a **mismatch**

A character aligned with a gap represents an **indel** (insertion/délétion)

—AC AC — GGTCTTA A T — —AATGGCC
CAG —G A A —G—AT— — C TTAGTTC — —

||| ||| ||| ||| A bar shows a mismatch

4 matches, 9 mismatches, 11 indels

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Basic algorithmic problem

Find an **alignment** of **two chains of** characters S and T that maximises

$\# \text{matches} - \# \text{mismatches} - \# \text{indels}$

The maximum value defines the **similarity** between sequences.

We speak of an "optimal global alignment". (We shall see more of this in SBAS, 2nd semester)

Biological interpretation: under the hypothesis of a **pointwise mutational model**, a mismatch represents a mutation, and an indel represents either an insertion or a deletion of a single character.

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Probabilities of insertion, deletion and substitution are used to construct alignments we look for: **substitution matrices**.

	ALA	ARG	ASN	ASP	CYS	GLN	GLU	GLY	HIS	ILE	LEU	LYS	MET	PHE	PRO	SER	THR	TRP	TYR	VAL
ALA	2	-2	0	0	-2	0	0	1	-1	-1	-2	-1	-1	-4	1	1	1	-6	-3	0
ARG	-2	6	0	-1	-4	1	-1	-3	2	-2	-3	3	0	-4	0	0	-1	2	-4	-2
ASN	0	0	2	2	-4	1	1	0	2	-2	-3	1	-2	-4	-1	1	0	-4	-2	-2
ASP	0	-1	2	4	-5	2	3	1	1	-2	-4	0	-3	-6	-1	0	0	-7	-4	-2
CYS	-2	-4	-4	-5	12	-5	-5	-3	-3	-2	-6	-5	-5	-4	-3	0	-2	-6	0	-2
GLN	0	1	1	2	-5	4	2	-1	3	-2	-2	1	-1	-5	0	-1	-1	-5	-4	-2
GLU	0	-1	1	3	-5	2	4	0	1	-2	-3	0	-2	-5	-1	0	0	-7	-4	-2
GLY	1	-3	0	1	-3	-1	0	5	-2	-3	-4	-2	-3	-5	-1	1	0	-7	-5	-1
HIS	-1	2	2	1	-3	3	1	-2	6	-2	-2	0	-2	-2	0	-1	-1	-3	0	-2
ILE	-1	-2	-2	-2	-2	-2	-3	-2	5	2	-2	2	1	-2	-1	0	-5	-1	4	-4
LEU	-2	-3	-3	-4	-6	-2	-3	-4	-2	2	6	-3	4	2	-3	-3	-2	-2	-1	2
LYS	-1	3	1	0	-5	1	0	-2	0	-2	-3	5	0	-5	-1	0	0	-5	-4	-2
MET	-1	0	-2	-3	-5	-1	-2	-3	-2	2	4	0	6	0	-2	-2	-1	-4	-2	2
PHE	-4	-4	-4	-6	-4	-5	-5	-2	1	2	-5	0	9	-5	-3	-3	0	7	-1	-1
PRO	1	0	-1	-1	-3	0	-1	-1	0	-2	-3	-1	-2	-5	6	1	0	-6	-5	-1
SER	1	0	1	0	0	-1	0	1	-1	-3	0	-2	-3	1	2	1	-2	-3	-1	-1
THR	1	-1	0	0	-2	-1	0	0	-1	0	-2	0	-1	-3	0	1	3	-5	-3	0
TRP	-6	2	-4	-7	-6	-5	-7	-7	-3	-5	-2	-3	-4	0	-6	-2	-5	17	0	-6
TYR	-3	-4	-2	-4	0	-4	-4	-5	0	-1	-1	-4	-2	7	-5	-3	-3	0	10	-2
VAL	0	-2	-2	-2	-2	-2	-2	-1	-2	4	2	-2	2	-1	-1	0	-6	-2	4	-4

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We define a cost/« penalty » for the set of possible operations.

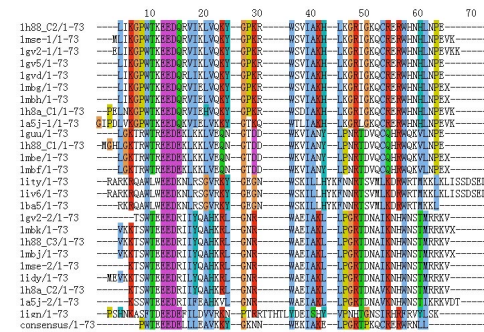
Example: insertion -> gap opening with penalty

Best alignment = alignment with the smallest sum of penalties

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The alignment can be among multiple sequences

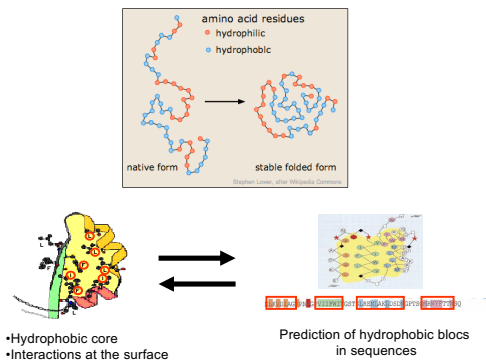


Physico-chemical properties are preserved

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Hydrophobicity, is the only physico-chemical property that is preserved in **distant homologs** (sequences with a common ancestor)



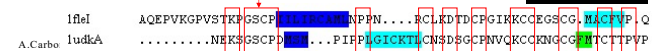
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Weak homologie: some difficult cases



Small blocs of identities



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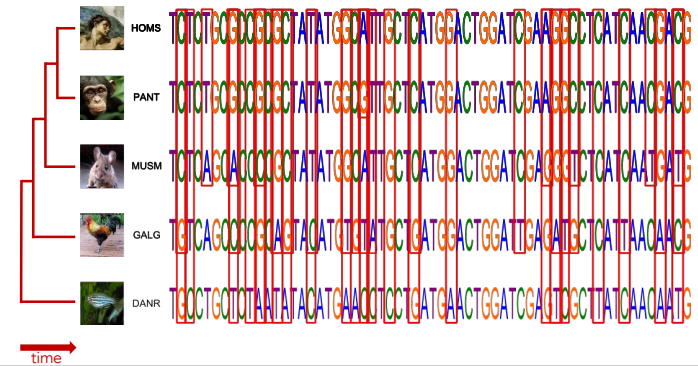
Let's take a moment to understand what is going on here:
sequences *evolve* in time, *they have a history*

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Proteins evolved with time

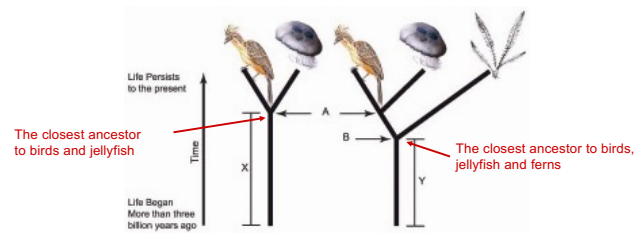


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Computational problem IV: reconstruction of phylogenetic trees

What are the genetic relations across species?

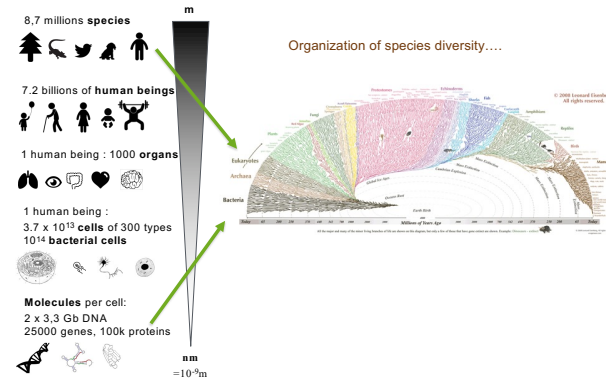
Idea : compare **specific characters** of different species, under the hypothesis that similar species are genetically close



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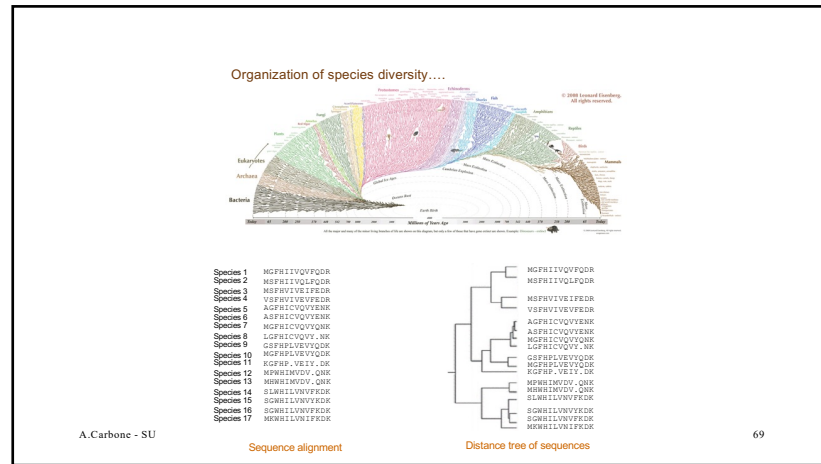
67



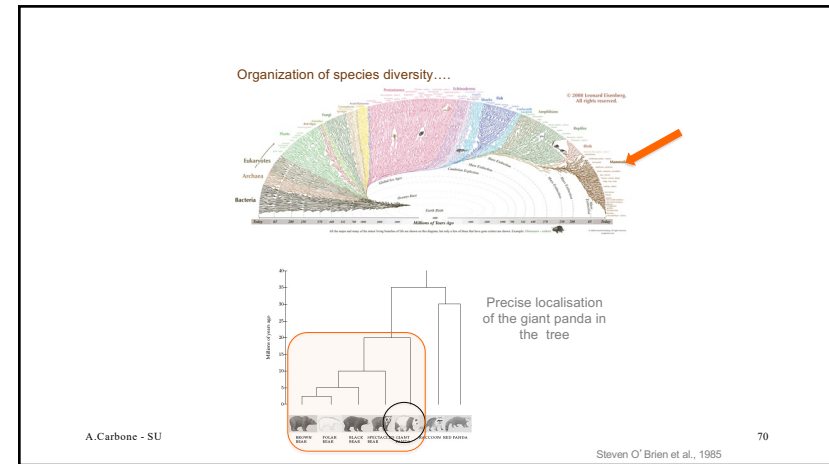
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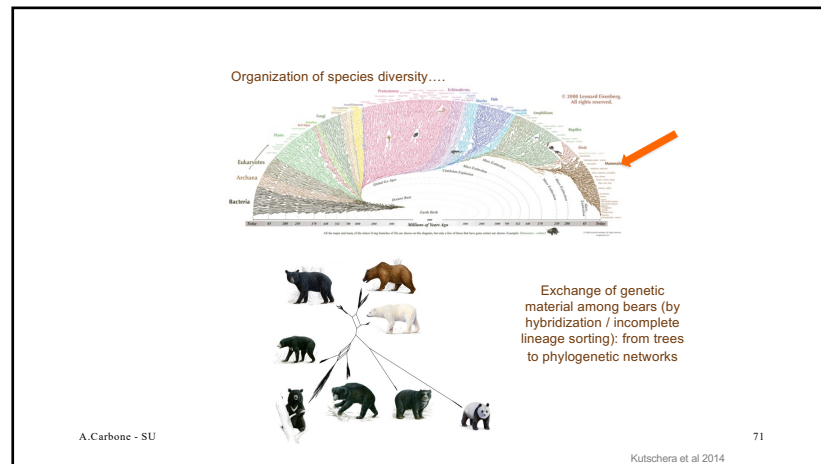
68



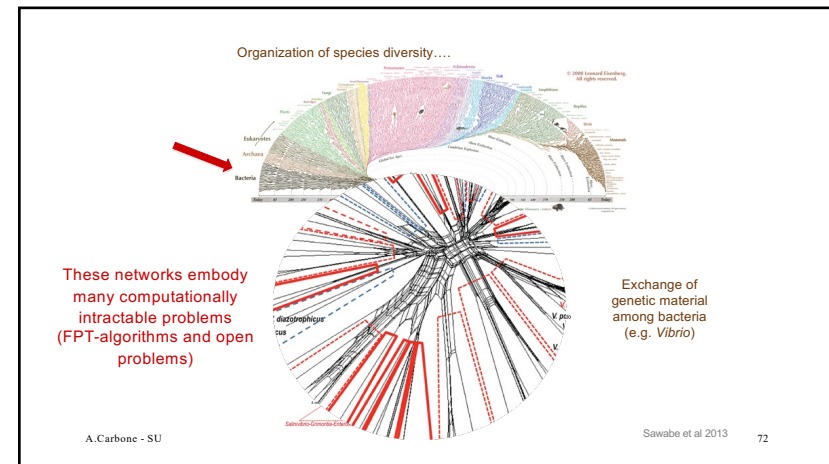
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Computational problem V :
structural mutations and genome rearrangements

Un exemple : human and mouse

- The mouse genome has 2.1×10^9 bp vs 2.9×10^9 bp for humans
- Approx 95% of the genetic material is shared.
- 99% of shared genes over about 20,000.
- 300 genes with no homologs in the two species concern immunity, detoxification, smell and sexuality.

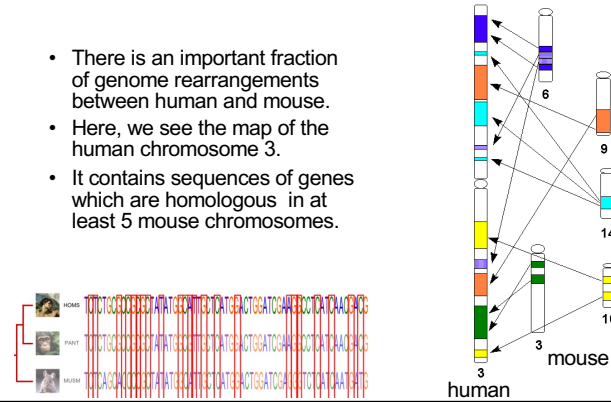
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Human and mouse

- There is an important fraction of genome rearrangements between human and mouse.
- Here, we see the map of the human chromosome 3.
- It contains sequences of genes which are homologous in at least 5 mouse chromosomes.

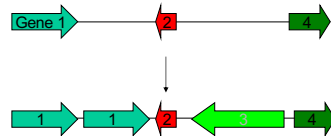


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74

Mechanisms:

- Gene duplication (one or several copies – gene 1)
- Insertions/deletion (gene 3)



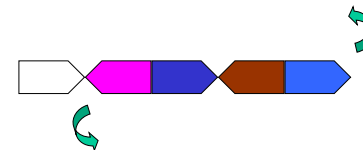
- Inversion of the genes...

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Comparison of the order of the genes:



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Comparison of the order of the genes:



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Comparison of the order of the genes:



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Comparison of the order of the genes:

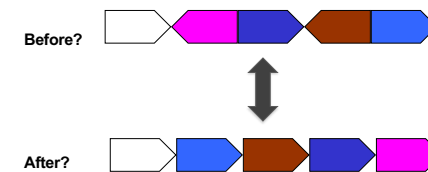


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Comparison of the order of the genes:



Evolution is present in the divergence of the **order of the genes**

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Evolution is also present in the divergence of the order of the genes. There are several mechanisms associated to rearrangements:

- Inversions at large scale
- Duplication of segments
- Transposition at large scale
- Fusion and fission
- Gene Horizontal Transfert
- Whole genome duplication



Genome rearrangements are rare events compared to pointwise mutations.

The sparsity of these events allows to establish the existence of evolutionary processes since the chance to “see” again the same rearrangement is extremely small. As a consequence, through the reconstruction of rearrangements, we can reconstruct evolutionary hypotheses.

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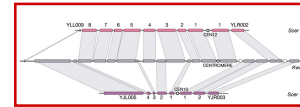
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Whole genome duplication for yeast

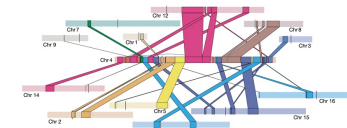
The budding yeast *S.cerevisiae* has extensive duplicates

We can trace a whole genome duplication by looking at yeast species that lack the duplicates (*K.waltii*, *A.gossipii*)

Only a small fraction (5%) of the yeast genome remains duplicated



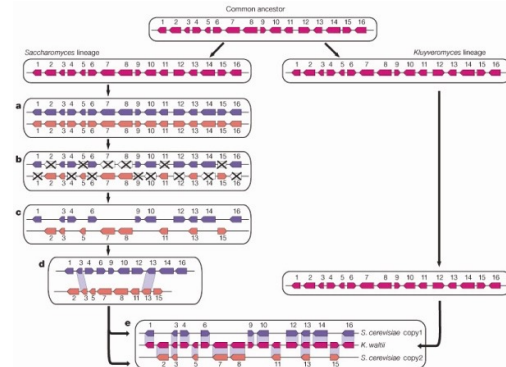
... and rearrangements



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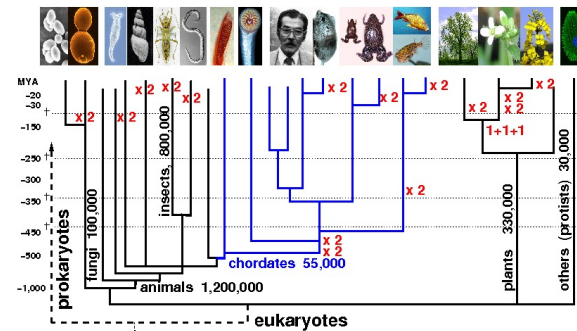
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Computational problem: given two permutations of a set of genomic segments, find the minimum number of operations to transform a permutation in the other.

Operations:

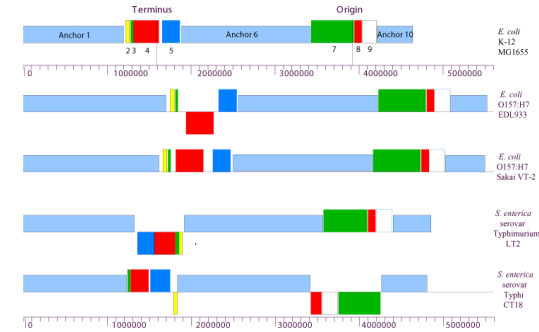
Inversions at large scale
Duplication of segments
Transposition at large scale
Fusion and fission

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Example 1: at a small scale, on a region of the genomes

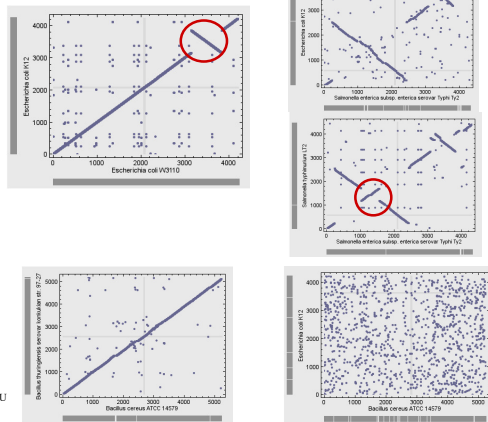
Rearrangements of Locally Collinear Blocks between three Escherichia and two Salmonella strains



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Example 2: at a large scale, at genome level

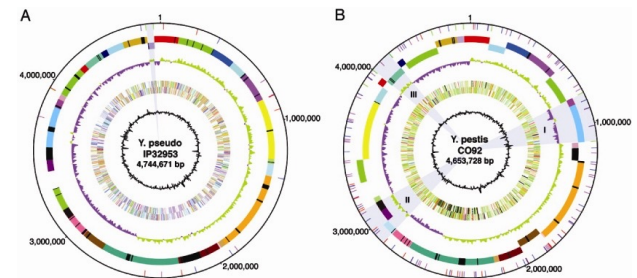


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Example 3: searching for differences

Rearrangement in bacteria :
Yersinia pseudotuberculosis versus *Yersinia pestis*



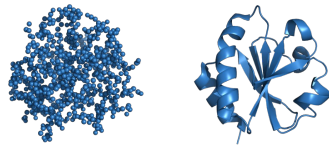
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Huge virulence

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Computational problem VI: protein folding

Given a protein sequence, predict the structure of the protein

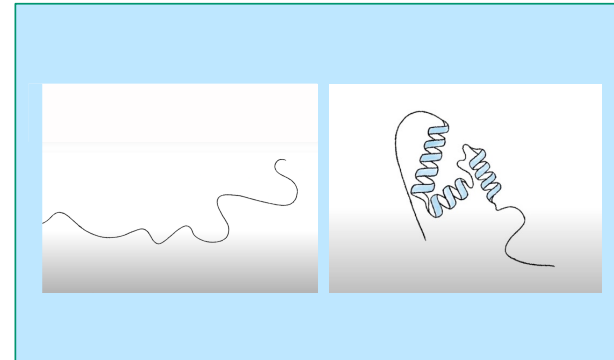


SDKIILHTDSDFDVLKADGAILVDFWAEJGCPCKMIAPILDEIADEYQGKLTVAKLINIDQNPSTAPKYGIRGIPTLLLFKNGEVAATKVGALSKGQLKEFLDANLA

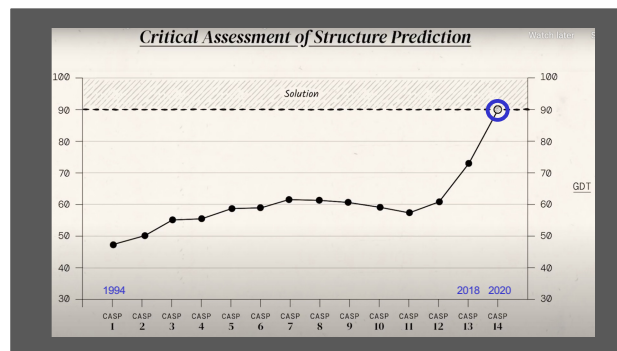
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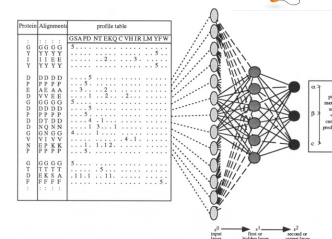


with Deep Learning!

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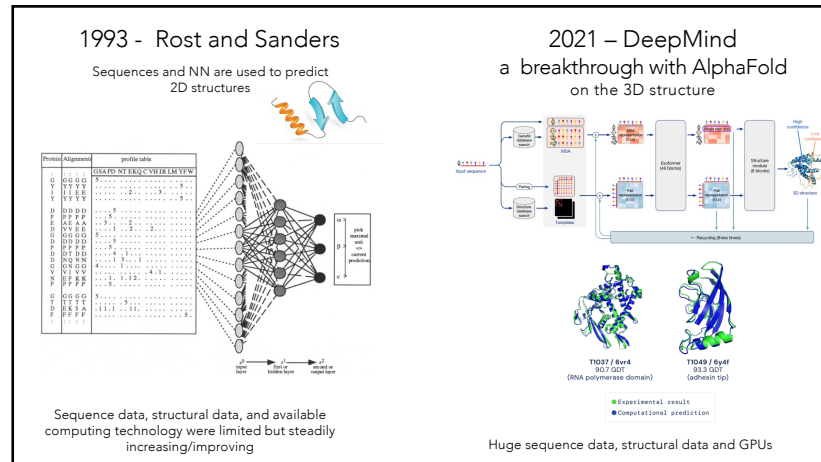
1993 - Rost and Sanders

Sequences and NN are used to predict
2D structures

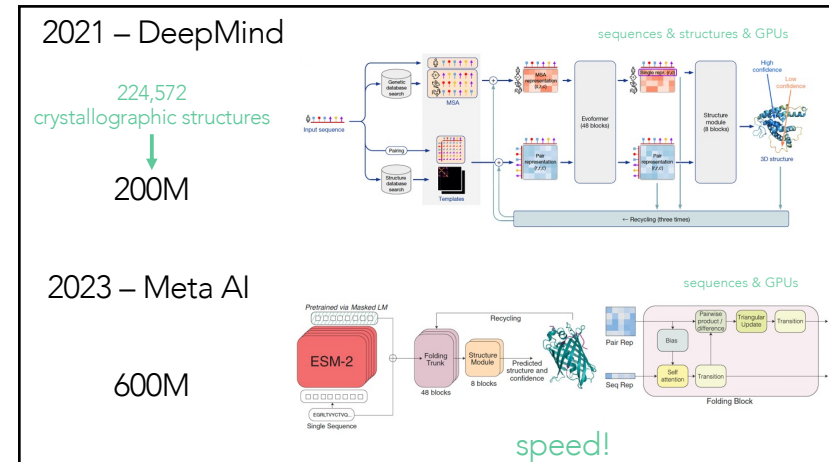


Sequence data, structural data, and available
computing technology were limited but steadily
increasing/improving

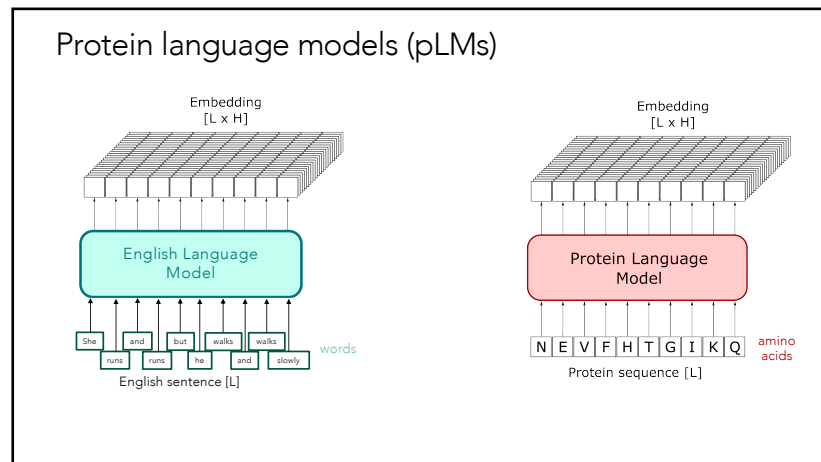
92



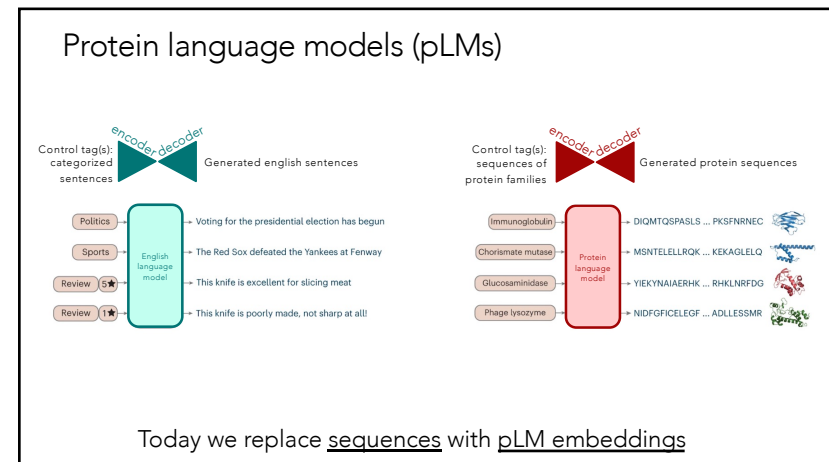
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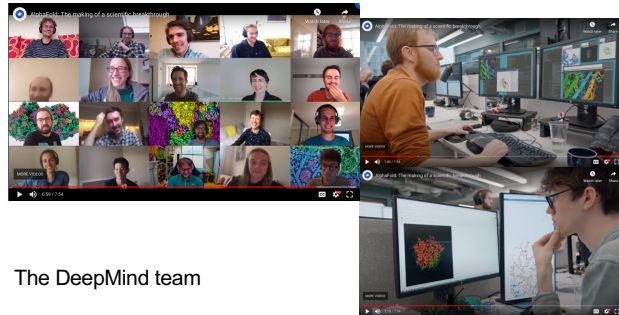
94



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The DeepMind team

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The physical model is missing but multiple information, going beyond our understanding, are probably identified by the neural networks, such as the potential conformations of the protein

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AI is revisiting today the encoding of information in genomic sequences and is using deep learning to uncover it **in a new and far more efficient way** than what our 'human eye' (mechanistic models, for example) can anticipate.

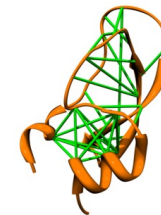
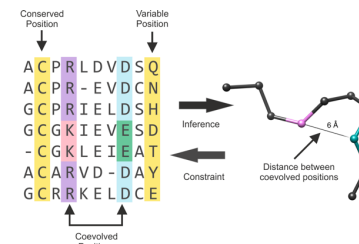
DNA and its encoding of biological information remind us that the information is hidden within billions (!!!!!) of sequences and their history in unexpected ways!

We are therefore obliged to take **a fresh look** at:

- Complexes
- Protein-protein/DNA, RNA/small molecule interactions
- The effects of genetic mutations

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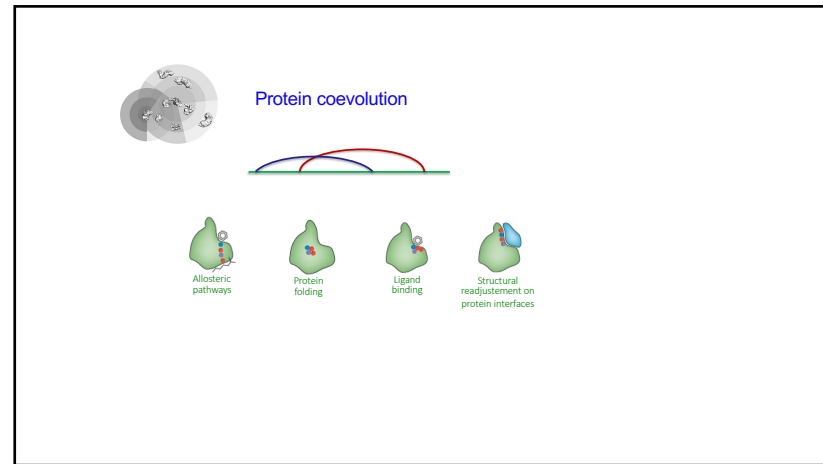
The crucial concept which is behind structural predictions



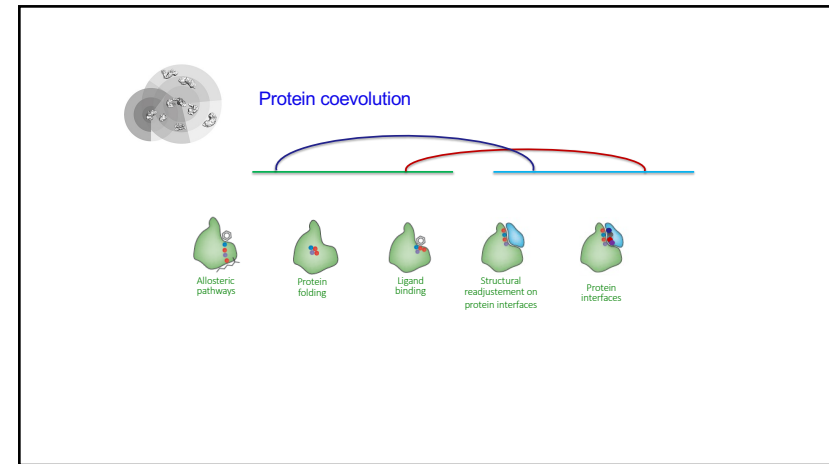
Top 30 DCA contacts depicted as green sticks in trypsin inhibitor (PDB ID: 5pti).

co-evolution in sequences

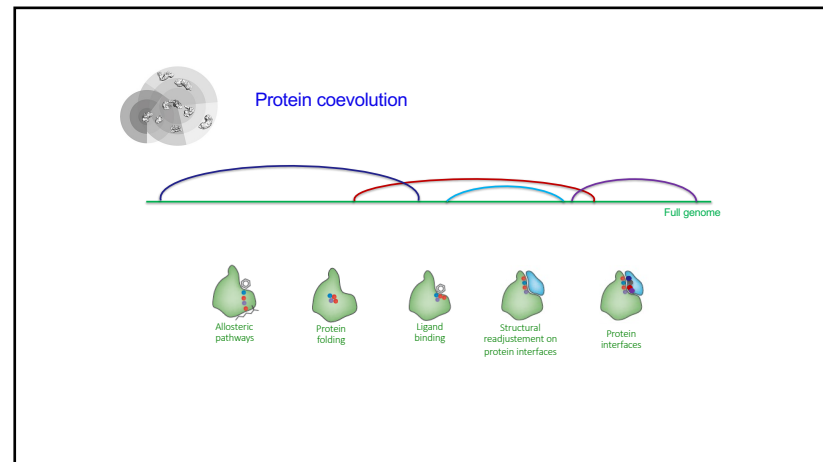
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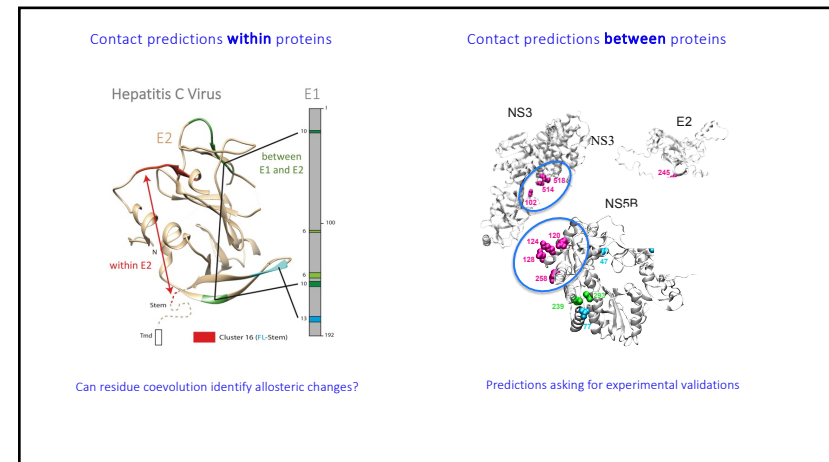
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102

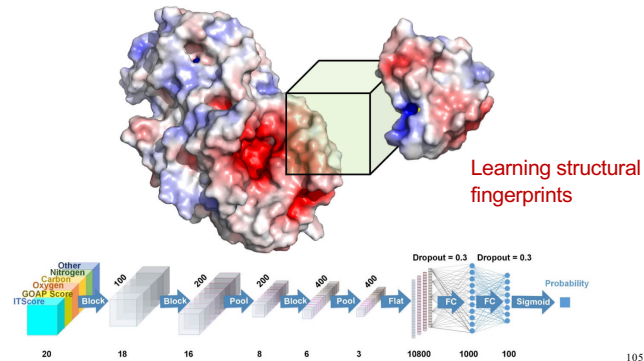


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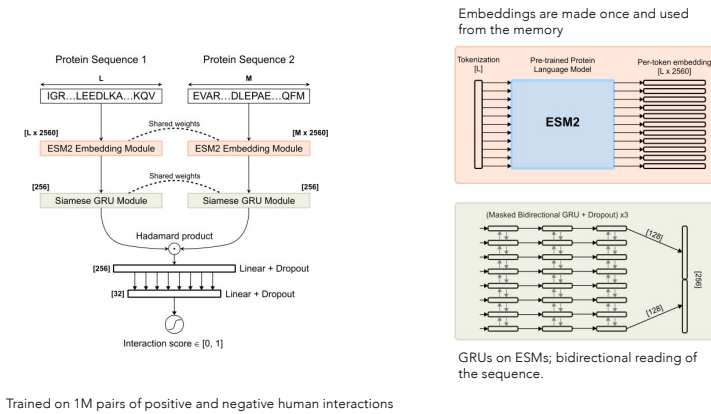
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Protein-protein interaction prediction using structures



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Protein-protein interaction prediction using sequences



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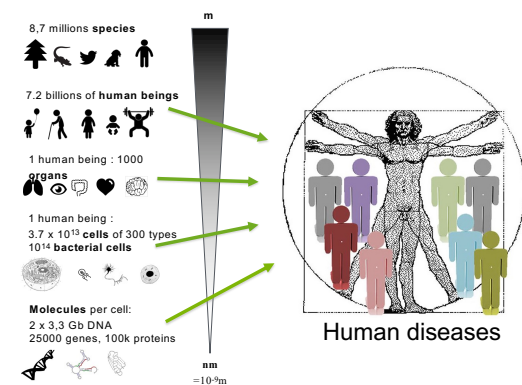
Computational problem VII: can we go from sequences to integrated behaviour

- Genome annotation
- Genome comparison
- Complex networks of interaction (metabolic, gene regulation, protein-protein interaction, protein-DNA interaction networks)
- Links between the different levels of the hierarchical organisation

Construction of omics databases

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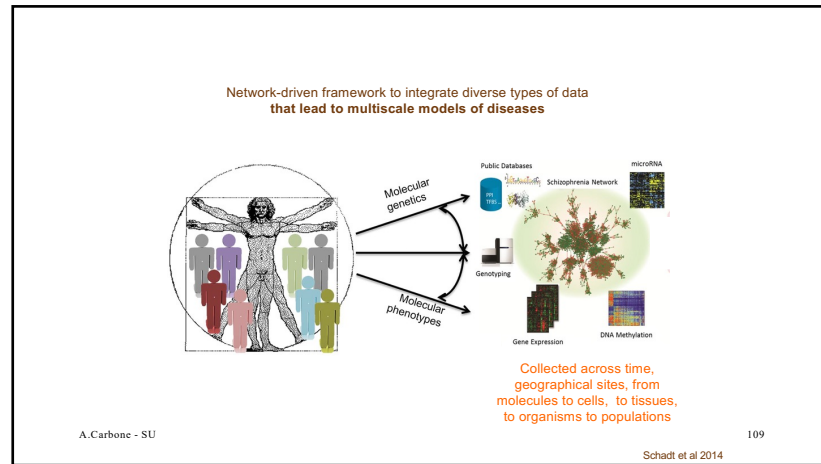
A.Carbonne - SU

Schadt et al 2014

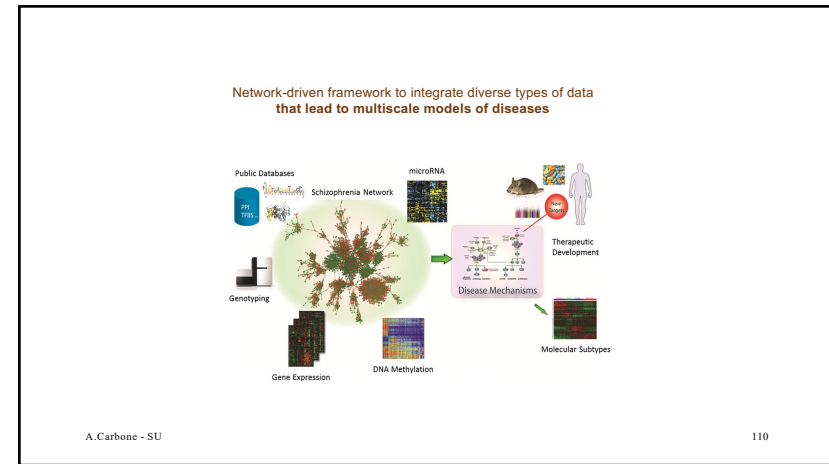
108

107

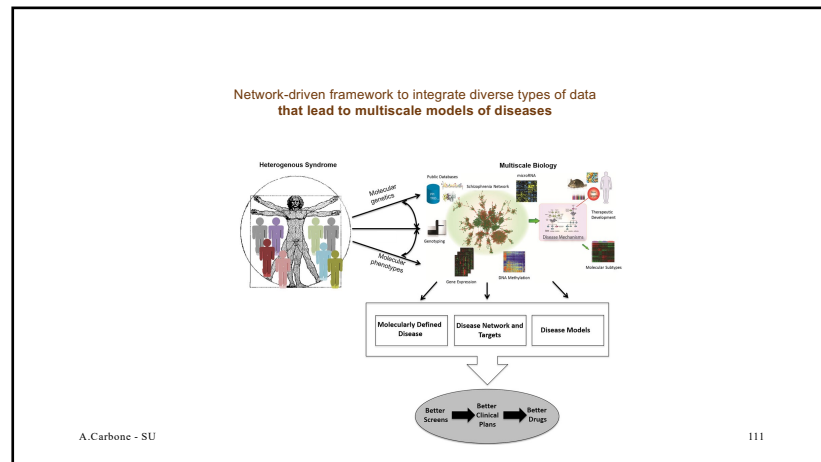
108



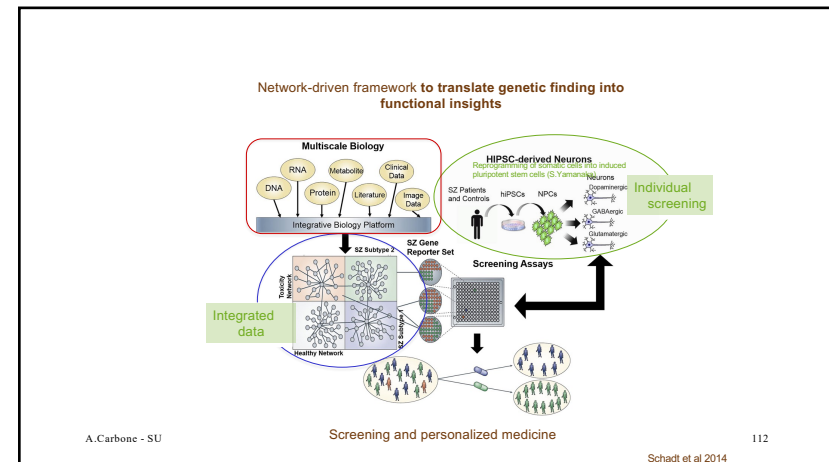
109



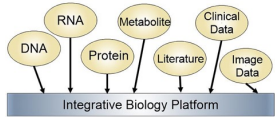
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Data-driven research

Accumulation of **data in high-throughput fashion** on a broad front (for instance, on all genomes) and available for sampling and interpretation.

Observational studies

Exploratory studies. Results might not be rigorously statistically supported but only suggestive, requiring confirmatory follow-up in experimental mode.

Hypothesis-driven research

Gathering of **data in a focused manner** to address a specific proposition, with a well-defined null hypothesis and all necessary controls

Experimental studies

Directed and well-controlled

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The development of techniques for the **effective integration of these two modes of investigation** is key to future progress in fields such as **molecular pharmacology**

Data-driven research

Accumulation of data in high-throughput fashion on a broad front (for instance, on all genomes) and available for sampling and interpretation.

Observational studies

Exploratory studies. Results might not be rigorously statistically supported but only suggestive, requiring confirmatory follow-up in experimental mode.

Hypothesis-driven research

Gathering of data in a focused manner to address a specific proposition, with a well-defined null hypothesis and all necessary controls

Experimental studies

Directed and well-controlled

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Homework:

Realise a thematic survey of existing databases in biology.

Choose a database and describe it carefully: structure, input, output. What would you like to see that is not already present?

To be returned to Gianluca LOMBARDI and Konstantin VOLZHENIN chargés de TD/TME after the fall break.

A. Carbone - SU

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